

**INTERNATIONAL CYBERSECURITY AND
DIGITAL FORENSICS ACADEMY**

Week 4 Practical Laboratory

Linux Security Monitoring and Auditing

From Technical Evidence to Governance Assurance

BY

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C11/26/CGRCE/17587

Evidence Bundle 1 – auditd Configuration and Events

1.1 – auditd service status

sudo systemctl status auditd

```
(icdfa@icdfa)-[~]
$ ls /mnt/c/Users/
ls: cannot access '/mnt/c/Users/': No such file or directory

(icdfa@icdfa)-[~]
$ sudo systemctl status auditd
[sudo] password for icdfa:
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-29 10:04:37 WAT; 19min ago
  Invocation: 4d7607757b8e45f188a4ea6424e3c3ad
     Docs: man:auditd(8)
          https://github.com/linux-audit/audit-documentation
 Main PID: 13599 (auditd)
    Tasks: 2 (limit: 2207)
  Memory: 1M (peak: 1.8M)
     CPU: 49ms
   CGroup: /system.slice/auditd.service
           └─13599 /usr/sbin/auditd

Sep 29 10:04:37 icdfa systemd[1]: Starting auditd.service - Security Audit Logging Service ...
Sep 29 10:04:37 icdfa auditd[13599]: No plugins found, not dispatching events
Sep 29 10:04:37 icdfa auditd[13599]: Init complete, auditd 4.1.2 listening for events (startup state enable)
Sep 29 10:04:37 icdfa systemd[1]: Started auditd.service - Security Audit Logging Service.

(icdfa@icdfa)-[~]
$
```

Fig 1.1 auditd service status

Evidence 1.2 – Loaded custom audit rules

```
File Actions Edit View Help

(icdfa@icdfa)-[~]
$ script ~/lab_evidence(2).txt
Script started, output log file is '/home/icdfa/lab_evidence(2).txt'.
(icdfa@icdfa)-[~]
$ sudo nano /etc/audit/rules.d/custom.rules
[sudo] password for icdfa:

(icdfa@icdfa)-[~]
$ sudo systemctl restart auditd

(icdfa@icdfa)-[~]
$ sudo auditctl -l
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -F key=program_execution
-a always,exit -F arch=b32 -S execve -F key=program_execution
-w /var/log/auth.log -p wa -k auth_failures

(icdfa@icdfa)-[~]
$
```

Fig 1.2 Loaded custom audit

The rules provide coverage for sensitive account files, program execution and authentication-log changes. The auth.log watch was loaded even though that traditional file was not present in the Kali environment

Evidence 1.3 – passwd_changes

sudo ausearch -k passwd_changes

```
(icdfa@icdfa)~$ sudo ausearch -k passwd_changes | tail -n 30
time-->Tue Sep 29 11:43:24 2026
type=PROCTITLE msg=audit(1790678604.781:14655): proctitle=7375646F006175736561726368002D68007061737377645F6368616E676573
type=PATH msg=audit(1790678604.781:14655): item=0 name="/etc/passwd" inode=1105587 dev=08:01 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790678604.781:14655): cwd="/home/icdfa"
type=SYSCALL msg=audit(1790678604.781:14655): arch=c000003e syscall=257 success=yes exit=0 a0=ffffff9c a1=7f0c52809d7f a2=80000 a3=0 items=1 ppid=3286 pid=25044 auid=1000 uid=1000 gid=1000 euid=0 suid=0 fsuid=0 egid=1000 fsgid=1000 tty=pts1 ses=2 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"

time-->Tue Sep 29 11:43:24 2026
type=PROCTITLE msg=audit(1790678604.781:14657): proctitle=7375646F006175736561726368002D68007061737377645F6368616E676573
type=PATH msg=audit(1790678604.781:14657): item=0 name="/etc/passwd" inode=1105587 dev=08:01 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790678604.781:14657): cwd="/home/icdfa"
type=SYSCALL msg=audit(1790678604.781:14657): arch=c000003e syscall=257 success=yes exit=13 a0=ffffff9c a1=7f0c52809d7f a2=80000 a3=0 items=1 ppid=3286 pid=25044 auid=1000 uid=1000 gid=1000 euid=0 suid=0 fsuid=0 egid=0 fsgid=0 tty=pts1 ses=2 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"
```

Fig 1.3 passwd_changes rules

Audit records were returned for activity involving /etc/passwd. The rule therefore generated evidence for access to the sensitive account file.

Evidence 1.4 – program_execution

sudo ausearch -k program_execution

```
(icdfa@icdfa)~$ sudo ausearch -k program_execution | tail -n 40
type=PATH msg=audit(1790678627.580:14802): item=0 name="/usr/bin/grep" inode=786639 dev=08:01 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790678627.580:14802): cwd="/home/icdfa"
type=EXECVE msg=audit(1790678627.580:14802): argc=4 a0="grep" a1="-o" a2="-P" a3=283F3C306966574202958302D395078312C337D285C2E58302D395078312C337D2978337D
type=SYSCALL msg=audit(1790678627.580:14802): arch=c000003e syscall=59 success=yes exit=0 a0=55fa37f480b0 a1=55fa37f47c38 a2=55fa37f47f58 a3=0 items=3 ppid=25232 pid=25234 auid=1000 uid=1000 gid=1000 euid=1000 fsuid=1000 egid=1000 fsgid=1000 tty=(none) ses=2 comm="grep" exe="/usr/bin/grep" subj=unconfined key="program_execution"

time-->Tue Sep 29 11:43:48 2026
type=PROCTITLE msg=audit(1790678628.180:14803): proctitle=7375646F006175736561726368002D680070726F6772616D5F65786563757469666E
type=PATH msg=audit(1790678628.180:14803): item=2 name="/lib64/ld-linux-x86-64.so.2" inode=792912 dev=08:01 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1790678628.180:14803): item=1 name="/usr/bin/sudo" inode=812903 dev=08:01 mode=0104755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1790678628.180:14803): item=0 name="/usr/bin/sudo" inode=812903 dev=08:01 mode=0104755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790678628.180:14803): cwd="/home/icdfa"
type=EXECVE msg=audit(1790678628.180:14803): argc=4 a0="sudo" a1="ausearch" a2="-k" a3="program_execution"
type=SYSCALL msg=audit(1790678628.180:14803): arch=c000003e syscall=59 success=yes exit=0 a0=7fdeb07fa40 a1=7f9c3b1120f8 a2=7fdeb0d8a5b0 a3=8 items=3 ppid=3286 pid=25235 auid=1000 uid=1000 gid=1000 euid=0 fsuid=0 fsgid=1000 tty=pts1 ses=2 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="program_execution"
```

Fig 1.4 program_execution

Records included executions associated with grep, sudo, tail, unix_chkpwd and ausearch. The execve rules were functioning and captured program execution.

Evidence 1.5 – aureport summary

sudo aureport

```
(icdfa@icdfa)~$ sudo aureport

Summary Report
=====
Range of time in logs: 29/09/2026 10:04:37.483 - 29/09/2026 11:28:02.868
Selected time for report: 29/09/2026 10:04:37 - 29/09/2026 11:28:02.868
Number of changes in configuration: 14
Number of changes to accounts, groups, or roles: 0
Number of logins: 1
Number of failed logins: 0
Number of authentications: 4
Number of failed authentications: 1
Number of users: 4
Number of terminals: 13
Number of host names: 2
Number of executables: 26
Number of commands: 32
Number of files: 32
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 2
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 4
Number of process IDs: 8706
Number of events: 9101
```

Fig 1.5 aureport summary

Evidence 1.6 – failed authentication summary

sudo aureport --failed

```
(icdfa@icdfa)-[~]
$ sudo aureport --failed

Failed Summary Report
=====
Range of time in logs: 29/09/2026 10:04:37.483 - 29/09/2026 11:30:00.207
Selected time for report: 29/09/2026 10:04:37 - 29/09/2026 11:30:00.207
Number of changes in configuration: 0
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 1
Number of users: 2
Number of terminals: 3
Number of host names: 2
Number of executables: 3
Number of commands: 1
Number of files: 1
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 2
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 1
Number of process IDs: 3
Number of events: 4
```

Fig 1.6 failed authentication

Evidence 1.7 – login summary

sudo aureport --login

```
(icdfa@icdfa)-[~]
$ sudo aureport --login

Login Report
=====
# date time auid host term exe success event
=====
1. 29/09/2026 10:54:55 1000 icdfa /dev/tty7 /usr/sbin/lightdm yes 59

(icdfa@icdfa)-[~]
$
```

Fig 1.7 login summary summary

Evidence Bundle 2 – Linux Log Analysis

Evidence 2.1 – sudo journal

sudo journalctl

```
(icdfa@icdfa)-[~]
$ sudo journalctl | tail -30
Sep 29 12:16:38 icdfa kernel: 11:16:38.622809 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:16:43 icdfa kernel: 11:16:38.623006 SHCLX11 Shared Clipboard: Converting X11 format 'image/bmp' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:16:43 icdfa kernel: 11:16:43.625928 SHCLX11 Requesting data in format 0x2 from host failed with VERR_NO_DATA
Sep 29 12:16:43 icdfa kernel: 11:16:43.626158 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:16:48 icdfa kernel: 11:16:43.626357 SHCLX11 Shared Clipboard: Converting X11 format 'image/bmp' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:16:48 icdfa kernel: 11:16:48.633481 SHCLX11 Requesting data in format 0x2 from host failed with VERR_NO_DATA
Sep 29 12:16:48 icdfa kernel: 11:16:48.633733 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:16:53 icdfa kernel: 11:16:48.633984 SHCLX11 Shared Clipboard: Converting X11 format 'image/bmp' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:16:53 icdfa kernel: 11:16:53.644153 SHCLX11 Requesting data in format 0x2 from host failed with VERR_NO_DATA
Sep 29 12:16:53 icdfa kernel: 11:16:53.644970 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:16:58 icdfa kernel: 11:16:53.645215 SHCLX11 Shared Clipboard: Converting X11 format 'image/bmp' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:16:58 icdfa kernel: 11:16:58.645999 SHCLX11 Requesting data in format 0x2 from host failed with VERR_NO_DATA
Sep 29 12:16:58 icdfa kernel: 11:16:58.646207 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:17:01 icdfa CRON[41633]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Sep 29 12:17:01 icdfa CRON[41633]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
Sep 29 12:17:01 icdfa CRON[41633]: pam_unix(cron:session): session closed for user root
Sep 29 12:17:03 icdfa kernel: 11:16:58.646409 SHCLX11 Shared Clipboard: Converting X11 format 'image/bmp' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:17:03 icdfa kernel: 11:17:03.654005 SHCLX11 Requesting data in format 0x2 from host failed with VERR_NO_DATA
Sep 29 12:17:03 icdfa kernel: 11:17:03.654318 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:17:08 icdfa kernel: 11:17:03.654584 SHCLX11 Shared Clipboard: Converting X11 format 'image/bmp' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:17:08 icdfa kernel: 11:17:08.659227 SHCLX11 Requesting data in format 0x2 from host failed with VERR_NO_DATA
Sep 29 12:17:08 icdfa kernel: 11:17:08.659607 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:17:13 icdfa kernel: 11:17:08.662099 SHCLX11 Shared Clipboard: Converting X11 format 'image/bmp' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:17:13 icdfa kernel: 11:17:13.671447 SHCLX11 Requesting data in format 0x2 from host failed with VERR_NO_DATA
Sep 29 12:17:13 icdfa kernel: 11:17:13.671691 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:17:18 icdfa kernel: 11:17:13.672450 SHCLX11 Shared Clipboard: Converting X11 format 'image/bmp' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:17:18 icdfa kernel: 11:17:18.678992 SHCLX11 Requesting data in format 0x2 from host failed with VERR_NO_DATA
Sep 29 12:17:18 icdfa kernel: 11:17:18.679435 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:17:20 icdfa sudo[41806]: icdfa : TTY=pts/1 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/journalctl
Sep 29 12:17:20 icdfa sudo[41806]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
```

Fig 2.1 – sudo journal

Recent entries showed repeated VirtualBox shared-clipboard messages. These were not treated as direct evidence of malicious activity.

Evidence 2.2 – SSH journal review

sudo journalctl -u ssh

```
(icdfa@icdfa)-[~]
$ sudo journalctl -u ssh | tail -30
-- No entries --
```

Fig 2.2 SSH journal review

No journal entries were available for those queried service units

Evidence 2.3 – current-day review

sudo journalctl --since "today"

```
(icdfa@icdfa)-[~]
$ sudo journalctl --since "today" | tail -30
Sep 29 12:25:44 icdfa kernel: 11:25:44.372370 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:25:49 icdfa kernel: 11:25:44.372606 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:25:49 icdfa kernel: 11:25:49.382528 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:25:49 icdfa kernel: 11:25:49.382735 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:25:54 icdfa kernel: 11:25:49.382931 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:25:54 icdfa kernel: 11:25:54.385605 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:25:54 icdfa kernel: 11:25:54.385812 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:25:59 icdfa kernel: 11:25:54.386010 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:25:59 icdfa kernel: 11:25:59.394307 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:25:59 icdfa kernel: 11:25:59.394602 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:26:04 icdfa kernel: 11:25:59.394874 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:26:04 icdfa kernel: 11:26:04.399358 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:26:04 icdfa kernel: 11:26:04.399721 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:26:09 icdfa kernel: 11:26:04.402016 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:26:09 icdfa kernel: 11:26:09.406221 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:26:09 icdfa kernel: 11:26:09.406492 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:26:14 icdfa kernel: 11:26:09.406752 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:26:14 icdfa kernel: 11:26:14.410761 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:26:14 icdfa kernel: 11:26:14.411608 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:26:19 icdfa kernel: 11:26:14.413953 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:26:19 icdfa kernel: 11:26:19.418103 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:26:19 icdfa kernel: 11:26:19.418408 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:26:24 icdfa kernel: 11:26:19.418719 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
```

Fig 2.3 current-day review

Recent current-day journal activity was displayed

Evidence 2.4 – error review

sudo journalctl -p err

```

(icdfa@icdfa)-[~]
$ sudo journalctl -p err
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupp
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics d
May 10 20:57:08 icdfa lightdm[761]: gkr-pam: unable to locate daemon control file
May 10 20:57:08 icdfa lightdm[645]: pam_systemd(lightdm-greeter:session): Failed to release session: Tra
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registry: Error calling StartServiceByName for org
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registry: Error calling StartServiceByName for org
-- Boot f8a18b2290a640c9b9236a0ff3f2a536 --
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupp
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics d
May 11 04:51:38 icdfa lightdm[802]: gkr-pam: unable to locate daemon control file
May 11 04:51:38 icdfa lightdm[690]: pam_systemd(lightdm-greeter:session): Failed to release session: Tra
May 11 04:51:48 icdfa obexd[1324]: Unable to acquire registry: Error calling StartServiceByName for org
May 11 04:51:48 icdfa obexd[1324]: Unable to acquire registry: Error calling StartServiceByName for org
-- Boot 1330474cb531439e929342180494b74a --
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupp
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics d
May 11 06:10:30 icdfa lightdm[785]: gkr-pam: unable to locate daemon control file
May 11 06:10:30 icdfa lightdm[678]: pam_systemd(lightdm-greeter:session): Failed to release session: Tra
May 11 06:10:42 icdfa obexd[1293]: Unable to acquire registry: Error calling StartServiceByName for org
May 11 06:10:42 icdfa obexd[1293]: Unable to acquire registry: Error calling StartServiceByName for org
-- Boot b8f421d015a94e7d82bf378d620f3ea1 --
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupp
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics d
May 11 06:23:02 icdfa lightdm[741]: gkr-pam: unable to locate daemon control file
May 11 06:23:02 icdfa lightdm[635]: pam_systemd(lightdm-greeter:session): Failed to release session: Tra
May 11 06:23:17 icdfa obexd[1256]: Unable to acquire registry: Error calling StartServiceByName for org
May 11 06:23:17 icdfa obexd[1256]: Unable to acquire registry: Error calling StartServiceByName for org
-- Boot 70de93df416a4f15a4e11555f8166985 --
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupp
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics d
May 11 06:30:16 icdfa lightdm[740]: gkr-pam: unable to locate daemon control file
May 11 06:30:16 icdfa lightdm[635]: pam_systemd(lightdm-greeter:session): Failed to release session: Tra
-- Boot ed33344071284c619cc899cd9af2cfed --
May 11 06:32:09 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupp
May 11 06:32:09 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 11 06:32:09 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics d
-- Boot 62145ff447bc4b8bbb86dbe7c56b0a9f --
May 11 06:37:32 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupp
May 11 06:37:32 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.

```

Fig 2.4 error review

Evidence 2.5 – live monitoring

sudo journalctl -f

```
└─$ sudo journalctl -f
Sep 29 12:34:44 icdfa kernel: 11:34:44.760310 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:34:44 icdfa kernel: 11:34:44.760521 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:34:49 icdfa kernel: 11:34:44.760715 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:34:49 icdfa kernel: 11:34:49.762200 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:34:49 icdfa kernel: 11:34:49.762419 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:34:54 icdfa kernel: 11:34:49.762661 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:34:54 icdfa kernel: 11:34:54.763805 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:34:54 icdfa kernel: 11:34:54.764398 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:34:56 icdfa sudo[50500]: icdfa : TTY=pts/1 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/
journalctl -f
Sep 29 12:34:56 icdfa sudo[50500]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(u
id=1000)
Sep 29 12:34:59 icdfa kernel: 11:34:54.765739 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:34:59 icdfa kernel: 11:34:59.768863 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:34:59 icdfa kernel: 11:34:59.769084 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:35:01 icdfa CRON[50553]: pam_unix(cron:session): session opened for user root(uid=0) by root(ui
d=0)
Sep 29 12:35:01 icdfa CRON[50555]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Sep 29 12:35:01 icdfa CRON[50553]: pam_unix(cron:session): session closed for user root
Sep 29 12:35:04 icdfa kernel: 11:34:59.769277 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:35:04 icdfa kernel: 11:35:04.770627 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:35:04 icdfa kernel: 11:35:04.770826 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
Sep 29 12:35:09 icdfa kernel: 11:35:04.771021 SHCLX11 Shared Clipboard: Converting X11 format 'image/bm
p' (idxFmtX11=10) to VBox format 0x2 failed, rc=VERR
Sep 29 12:35:09 icdfa kernel: 11:35:09.772052 SHCLX11 Requesting data in format 0x2 from host failed wi
th VERR_NO_DATA
Sep 29 12:35:09 icdfa kernel: 11:35:09.772252 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP
' to 'image/bmp' for X11 (idxFmtX11=10, fmtX11=4, at
^C
```

Fig 2.5 live monitoring

Evidence 2.6 – privilege-use review

sudo journalctl | grep -i "sudo"

```
(icdfa@icdfa)-[~]
$ sudo journalctl | grep -i "sudo"
May 10 21:20:11 icdfa sudo[13196]: pam_unix(sudo:auth): authentication failure; logname=icdfa uid=1000 euid=0 tty=/dev/pts/0 ruser=icdfa rhost= user=icdfa
May 10 21:20:17 icdfa sudo[13196]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/apt -y update
May 10 21:20:17 icdfa sudo[13196]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
May 10 21:30:02 icdfa sudo[13196]: pam_unix(sudo:session): session closed for user root
May 10 21:30:02 icdfa sudo[18007]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/dpkg --add-architecture i386
May 10 21:30:02 icdfa sudo[18007]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
May 10 21:30:02 icdfa sudo[18007]: pam_unix(sudo:session): session closed for user root
May 10 21:30:02 icdfa sudo[18011]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/apt -y update
May 10 21:30:02 icdfa sudo[18011]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
May 10 21:38:46 icdfa sudo[18011]: pam_unix(sudo:session): session closed for user root
May 10 21:38:46 icdfa sudo[22246]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/apt -y install wine64 wine32
May 10 21:38:46 icdfa sudo[22246]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
May 11 00:03:09 icdfa lynis[95911]: - Sudoers file(s) [ FOUND ]
May 11 00:03:09 icdfa lynis[95911]: - Permissions for directory: /etc/sudoers.d [ WARNING ]
May 11 00:03:09 icdfa lynis[95911]: - Permissions for: /etc/sudoers [ OK ]
May 11 00:03:09 icdfa lynis[95911]: - Permissions for: /etc/sudoers.d/kali-grant-root [ OK ]
May 11 00:03:09 icdfa lynis[95911]: - Permissions for: /etc/sudoers.d/README [ OK ]
May 11 04:55:10 icdfa sudo[3034]: icdfa : TTY=pts/0 ; PWD=/home/icdfa/tools ; USER=root ; COMMAND=./1.sh
May 11 04:55:10 icdfa sudo[3034]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
May 11 04:55:10 icdfa sudo[3066]: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/apt -y update
May 11 04:55:10 icdfa sudo[3066]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=0)
May 11 04:55:18 icdfa sudo[3066]: pam_unix(sudo:session): session closed for user root
May 11 04:55:18 icdfa sudo[3165]: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/dpkg --add-architecture i386
May 11 04:55:18 icdfa sudo[3165]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=0)
May 11 04:55:18 icdfa sudo[3165]: pam_unix(sudo:session): session closed for user root
May 11 04:55:18 icdfa sudo[3169]: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/apt -y update
May 11 04:55:18 icdfa sudo[3169]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=0)
May 11 04:55:22 icdfa sudo[3169]: pam_unix(sudo:session): session closed for user root
May 11 04:55:22 icdfa sudo[3220]: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/apt -y install wine64 wine32
May 11 04:55:22 icdfa sudo[3220]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=0)
May 11 04:55:22 icdfa sudo[3220]: pam_unix(sudo:session): session closed for user root
May 11 04:55:22 icdfa sudo[3225]: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/apt -y install wine
May 11 04:55:22 icdfa sudo[3225]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=0)
May 11 04:55:22 icdfa sudo[3225]: pam_unix(sudo:session): session closed for user root
May 11 04:55:22 icdfa sudo[3241]: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/apt -y install python3-pip
May 11 04:55:22 icdfa sudo[3241]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=0)
May 11 04:55:22 icdfa sudo[3241]: pam_unix(sudo:session): session closed for user root
May 11 04:55:22 icdfa sudo[3249]: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/apt -y install leafpad
May 11 04:55:22 icdfa sudo[3249]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=0)
May 11 04:55:22 icdfa sudo[3249]: pam_unix(sudo:session): session closed for user root
```

Fig 2.6 privilege-use review

Evidence 2.7 – failed-password search

sudo journalctl | grep -i "failed password"

```
(icdfa@icdfa)-[~]
$ sudo journalctl | grep -i "failed password"
Sep 29 12:41:23 icdfa sudo[53870]: icdfa : TTY=pts/1 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/grep -i 'failed password' /var/log/auth.log
```

Fig 2.7 failed-password search

Evidence 2.8 – warning review

sudo journalctl -p warning

```
(icdfa@icdfa)-[~]
$ sudo journalctl -p warning
May 10 20:50:54 icdfa kernel: acpi PNP0A03:00: fail to add MMCONFIG information, can't access extended configuration space under this bridge
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
May 10 20:50:54 icdfa systemd[1]: Failed to read SMBIOS type #11 object 0, ignoring: Bad message
May 10 20:50:54 icdfa systemd-sysv-generator[339]: SysV service '/etc/init.d/inetd' lacks a native systemd unit file, automatically generating a unit file for compatibility.
May 10 20:50:54 icdfa systemd-sysv-generator[339]: Please update package to include a native systemd unit file.
May 10 20:50:54 icdfa systemd-sysv-generator[339]: ! This compatibility logic is deprecated, expect removal soon. !
May 10 20:50:57 icdfa (haveged)[503]: haveged.service: Referenced but unset environment variable evaluates to an empty string: DAEMON_ARGS
May 10 20:51:02 icdfa lightdm[623]: Could not enumerate user data directory /var/lib/lightdm/data: Error opening directory '/var/lib/lightdm/data': No such file or directory
May 10 20:51:11 icdfa wireplumber[677]: spa.alsa: The decibel volume range for element 'Center' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:51:11 icdfa wireplumber[677]: spa.alsa: The decibel volume range for element 'LFE' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:51:11 icdfa wireplumber[677]: spa.alsa: The decibel volume range for element 'Center' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:51:11 icdfa wireplumber[677]: spa.alsa: The decibel volume range for element 'LFE' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:51:11 icdfa wireplumber[677]: spa.alsa: The decibel volume range for element 'Center' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:51:11 icdfa wireplumber[677]: spa.alsa: The decibel volume range for element 'LFE' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:51:11 icdfa wireplumber[677]: spa.alsa: The decibel volume range for element 'Center' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:51:11 icdfa wireplumber[677]: spa.alsa: The decibel volume range for element 'LFE' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:51:12 icdfa wireplumber[677]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
May 10 20:51:12 icdfa wireplumber[677]: spa.bluez5: BlueZ system service is not available
May 10 20:57:08 icdfa lightdm[761]: gdm-pam: unable to locate daemon control file
May 10 20:57:08 icdfa lightdm[645]: pam_systemd(lightdm-greeter:session): Failed to release session: Transport endpoint is not connected
May 10 20:57:09 icdfa wireplumber[798]: spa.alsa: The decibel volume range for element 'Center' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:57:09 icdfa wireplumber[798]: spa.alsa: The decibel volume range for element 'LFE' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:57:09 icdfa wireplumber[798]: spa.alsa: The decibel volume range for element 'Center' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:57:09 icdfa wireplumber[798]: spa.alsa: The decibel volume range for element 'LFE' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:57:09 icdfa wireplumber[798]: spa.alsa: The decibel volume range for element 'Center' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:57:09 icdfa wireplumber[798]: spa.alsa: The decibel volume range for element 'LFE' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:57:09 icdfa wireplumber[798]: spa.alsa: The decibel volume range for element 'Center' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:57:09 icdfa wireplumber[798]: spa.alsa: The decibel volume range for element 'LFE' (-4650 dB - -2400 dB) has negative maximum. Disabling the decibel range.
May 10 20:57:09 icdfa wireplumber[798]: default: Failed to get percentage from UPower: org.freedesktop.DBus.Error.NameHasNoOwner
May 10 20:57:09 icdfa wireplumber[798]: spa.bluez5: BlueZ system service is not available
May 10 20:57:15 icdfa colord[1072]: failed to get edid data: EDID length is too small
May 10 20:57:18 icdfa systemd[652]: mpris-proxy.service: Failed with result 'exit-code'.
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-registry.service not found
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-registry.service not found
May 10 21:18:30 icdfa org.freedesktop.portal.Documents[12275]: Ignoring invalid max threads value 4294967295 > max (100000).
```

Fig 2.8 warning review

Module 2 Event Table

Event/Condition	Evidence Source	Time	User/Service	Security Significance	Recommended Action
VirtualBox clipboard events	journalctl	29 Sep 2026 recent review	icdfa/VirtualBox	VM integration messages; not direct attack evidence	Monitor/investigate if persistent
No SSH journal entries	journalctl -u ssh/sshd	29 Sep 2026	SSH service units	No entries available from queried units	Confirm SSH status separately if in scope
Sudo authentication failure	journalctl grep sudo	10 May 21:20:11	icdfa/sudo	Privileged authentication failure	Review and monitor repeated occurrences
Successful privileged session	journalctl grep sudo	10 May 21:20:17	icdfa/sudo	Root-level activity followed the failure	Correlate user, command and authorization
vmwgfx errors	journalctl -p err	10 May 20:50:54	kernel/vmwgfx	VM configuration issue	Review VM configuration if persistent
System warnings	journalctl -p warning	10 May 2026	icdfa/systemd	Configuration/compatibility conditions	Monitor recurring warnings

Evidence Bundle 3 – Lynis Security Assessment

Evidence 3.1 – Lynis audit summary

sudo lynis audit system

```
Lynis security scan details:

Hardening index : 65 [##### ]
Tests performed : 273
Plugins enabled : 1

Components:
- Firewall [V]
- Malware scanner [V]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

Lynis 3.1.4

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2024, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
```

Fig 3.1 Lynis audit summary

```
(icdfa@icdfa)-[~]
$ lynis --version
3.1.4

(icdfa@icdfa)-[~]
$
```

Lynis version

3.1 Selected Lynis Findings

Finding	Type	Risk / Control Issue	Owner	Priority / Action
FIRE-4512 – iptables modules loaded, no rules active	Warning	Validate approved firewall architecture; possible filtering gap if no alternative exists	Linux/System Administrator; Security/Network Security	Immediate validation; remediate if control absent
DEB-0880 – Install Fail2ban	Suggestion	Potential additional protection against repeated authentication errors	Security Operations / System Administrator	Assess applicability
BOOT-5122 – GRUB password	Suggestion	Additional boot-configuration protection	Linux/System Administrator / Security	Assess against baseline
AUTH-9282/AUTH-9286 – password age controls	Suggestion	Authentication settings should align with policy	IAM/System Administrator	Review against policy
LOGG-2154 – external logging	Suggestion	Centralized logging improves retention/monitoring resilience	Security Operations / SIEM Administrator	Assess logging architecture
FILE-7524/HOME-9304 – permissions	Suggestion	Potential least-privilege weakness	Linux/System Administrator	Review identified permissions

EVIDENCE BUNDLE 4: Control Monitoring and Governance

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
Firewall/network filtering	Lynis FIRE-4512	Linux/System Administrator; Security/Network Security	Requires validation	Required firewall active	Possible network exposure	Confirm firewall technology; implement required filtering	Verify status/rules and rerun Lynis
Audit logging	auditd active; custom rules loaded	Linux Administrator / Security-GRC	Operating at test time	auditd active; required rules present	Loss of audit evidence	Monitor service/rules	Check service, auditctl -l and events

Privileged access	sudo failure followed by successful root activity	System Administrator / IAM / Security	Requires monitoring	Repeated/unexplained failures trigger review	Potential access-control concern	Review context/correlate activity	Review subsequent logs
Authentication protection	Lynis DEB-0880	Security Operations / System Administrator	Improvement opportunity	Failures above approved threshold	Possible password-guessing exposure	Assess Fail2ban/equivalent	Controlled test
Boot security	Lynis BOOT-5122	Linux Administrator / Security	Hardening recommendation	Required boot protection where applicable	Unauthorized boot-parameter changes	Assess/configure if required	Verify config and normal boot
Password controls	Lynis AUTH-9282/9286	IAM/System Administrator	Requires review	Meet approved password policy	Reduced account protection	Review login configuration	Verify settings/Lynis
Centralized logging	Lynis LOGG-2154	Security Operations / SIEM Administrator	Improvement opportunity	Critical logs reach central platform	Reduced retention/resilience	Implement approved forwarding	Test event centrally
File permissions	Lynis FILE-7524/HOME-9304	Linux Administrator	Requires review	Meet least-privilege baseline	Unauthorized access risk	Review/correct permissions	Recheck and rerun assessment

Evidence Bundle 5 – SIEM, Automation and Continuous Monitoring

Linux Evidence	Monitoring Objective	SIEM Representation	Possible Alert	Governance Significance
auditd file events	Sensitive-file monitoring	Host/user/time/action fields	Unauthorized/unusual access	Potential file-integrity/access-control issue
auditd execve	Command monitoring	User/command/host/time	Defined sensitive commands	Issue if privileged-use control is breached

sudo/auth events	Privileged access	User/result/target/comm and	Repeated failures or unexplained success	Escalate when thresholds are breached
auditd status/rules	Audit assurance	Scheduled control check	auditd/rules inactive	Control failure due to loss of audit coverage
journal errors/warnings	Operational monitoring	Service/severity/time	Severity/repetition threshold	Governance only if control affected
Lynis findings	Security configuration	Test ID/finding/remediation state	Open/high-impact findings	Control issue if mandatory baseline not met
Firewall status	Network control	Approved firewall/ruleset	Required filtering inactive	Potential governance issue
Central logging	Logging control	Ingestion/source status	Log-source silence/failure	Control issue if critical logs are not retained

Enterprise flow: Linux host → auditd/journal/Lynis → collection/integration → SIEM normalization → correlation/analytics → alert → investigation → GRC finding where applicable → remediation → retest → closure.

7.1 Technical Alert vs Governance Issue

Condition	Initial Treatment	Governance Escalation
VirtualBox/VM graphics messages	Technical operational alert	Only if shown to affect a defined security/availability control
Single sudo failure	Security monitoring event	If repeated, unexplained or unauthorized
auditd active/rules loaded	Control operating evidence	Escalate if auditd/rules become inactive
Lynis suggestion	Hardening opportunity	When a mandatory baseline/policy control is not met

FIRE-4512	Validate technical configuration	If required firewall control is absent/ineffective
Centralized logging gap	Control-assurance concern	If critical logs required by policy are not forwarded/retained

Governance Escalation Questions

Highest Priority Finding

The FIRE-4512 firewall warning should receive immediate technical validation because it explicitly states that iptables modules were loaded but no active rules were detected. Network filtering is an important preventive control. However, this must first be checked against the actual firewall architecture because another approved firewall technology may be operating. If the required control is absent, it should be treated as a significant control weakness.

Remediation Ownership

Finding	Primary Owner	Supporting Function
Firewall	Linux/System Administrator	Security/Network Security
Auditd/rules	Linux/System Administrator	Security/GRC
Privileged access	System Administrator / IAM	Security Operations
Authentication protection	Security Operations	IAM/System Administrator
GRUB	Linux/System Administrator	Security
Password controls	IAM/System Administrator	Security/GRC
Centralized logging	Security Operations / SIEM Administrator	Infrastructure
File permissions	Linux/System Administrator	Security/GRC

Escalation Thresholds

- Repeated authentication failures exceed the approved threshold.

- An unexplained successful privileged login follows repeated failures.
- Unauthorized privileged activity is identified.
- auditd becomes inactive or required rules are removed without authorization.
- Critical security logs are unavailable beyond the approved tolerance.
- Required firewall protection is inactive or unauthorized firewall changes occur.
- A mandatory baseline finding remains unresolved beyond its approved remediation deadline.
- A finding exceeds risk appetite or has significant business, regulatory or security impact.

The laboratory does not specify universal numeric thresholds. Exact values should be defined by the organization's security policy, risk appetite and control owners.

Evidence of Successful Remediation

- Firewall: approved configuration, active rules/status and follow-up assessment.
- Auditd: active service, required rules and new audit events.
- Privileged access: reviewed logs and confirmation of authorized activity.
- Authentication protection: configuration and controlled test evidence.
- GRUB: authentication configuration and successful boot.
- Password controls: configuration aligned with policy.
- Centralized logging: test event visible in the central platform.
- File permissions: before/after permission evidence and follow-up assessment.

Retesting

Retest after remediation and periodically according to risk. Record the baseline, implement the approved fix, verify the configuration, generate/observe a test event where appropriate, run the technical check again, compare with the baseline, document the evidence and close or escalate the finding.

Control	Suggested Retest
Auditd	Continuous monitoring with periodic formal verification
Privileged access	Continuous monitoring with periodic access review
Authentication	Continuous monitoring
Firewall	After significant changes and during scheduled reviews
Password controls	After changes and periodic reviews
GRUB	After implementation and during baseline reviews
Centralized logging	Continuous monitoring with periodic verification
File permissions	After remediation and during security assessments
Lynis	Periodically and after significant configuration changes

Findings and Remediation Register

ID	Evidence	Finding	Owner	Priority	Recommended Action	Closure Evidence
W4-F01	Lynis FIRE-4512	No active iptables rules detected	Linux Admin/Security	High pending validation	Validate firewall architecture; remediate if required	Firewall status/rules + Lynis retest
W4-F02	journalctl sudo	Authentication failure followed by successful privileged activity	System Admin/IAM/Security	Moderate pending context	Review and correlate activity	Reviewed privilege/auth logs
W4-F03	auditd	Service active and rules loaded	Linux Admin/Security	Operating control	Maintain continuous monitoring	Service/rules/audit-event evidence
W4-F04	Lynis DEB-0880	Fail2ban recommended	Security Ops/System Admin	Moderate	Assess applicability and implement if required	Configuration + controlled test
W4-F05	Lynis LOGG-2154	External logging recommended	Security Ops/SIEM Admin	Moderate	Assess and implement central	Central test event

					forwarding if required	
--	--	--	--	--	---------------------------	--

Conclusion

The laboratory shows how Linux technical evidence can support security governance and control assurance. auditd was active and custom rules were loaded, while ausearch and aureport produced evidence of file activity, command execution and authentication events.

journalctl provided system, authentication and privilege-use evidence. The genuine sudo authentication failure followed by successful privileged activity is relevant to access monitoring, but should not be treated as proof of malicious activity without further context.

Lynis 3.1.4 performed 273 tests and produced a hardening index of 65. The FIRE-4512 firewall warning requires validation against the actual firewall architecture. Other recommendations provide opportunities to strengthen authentication, boot security, password management, logging and permissions.

The evidence can be extended into enterprise monitoring through SIEM collection, normalization, correlation, alerting, dashboards and automated control checks. A technical event becomes a governance issue when it demonstrates that an approved control is absent, ineffective, unauthorized or outside the defined risk tolerance.